WEBINAR

The Blind Spots in Fraud Models

You fixed your model so it stops missing known fraud. What about the fraud it's never seen?

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SECTION 01

The Hook

THE SETUP

10,000 Loan Applicants. One Question: Who Defaults?

A loan portfolio, like most fraud and credit risk problems, is heavily imbalanced. Almost everyone pays.

9,400

Good loans
94% of the portfolio

600

Defaults
6% of the portfolio

94 : 6

that ratio is the entire problem

THE NAIVE MODEL

Predict “good loan” for every single applicant.
No features. No machine learning. Just a guess.

94%

accuracy

0%

defaults caught

Same model. Two numbers. Only one of them tells you the truth.

THE CONFUSION MATRIX DOESN'T LIE

Where Did the 600 Defaults Go?

9,400 TRUE NEGATIVE Good loan, correctly approved	**0** FALSE POSITIVE Good loan, wrongly declined
600 FALSE NEGATIVE Default, missed entirely	**0** TRUE POSITIVE Default, correctly caught

“That’s not a model.
That’s a liability
dressed up as a metric.”

SECTION 02

The Standard Fix

THE PLAYBOOK

Three Fixes, In the Order You Should Try Them

TRY FIRST · FREE

Class Weights

One parameter. Tells the model a missed default costs more than a false alarm. Zero data manipulation.

TRY SECOND · TUNE

Threshold Tuning

0.5 is an assumption, not a law. Move the decision line to where recall and precision actually balance for your business.

LAST RESORT

SMOTE

Synthesize new minority-class samples. Helpful, not magic — and only after the first two fixes are already in place.

THE CODE, BRIEFLY

This Is a One-Line Fix

```
model = GradientBoostingClassifier(random_state=42)
model.fit(X_train, y_train, sample_weight=compute_sample_weight(
    class_weight=
"balanced", y=y_train))
```

WHY IT WORKS

Class weights scale the loss function so errors on the minority class are penalized more heavily. The model sees 600 defaults as if they were roughly 9,400 — without adding or removing a single row of data.

THE NUMBERS

Before vs. After, Same Test Set

Model	Recall	Precision	F1	AUC
Baseline (no fix)	12%	58%	0.20	0.891
Class weights	61%	53%	0.56	0.901
+ Threshold tuning	**78%**	**61%**	**0.68**	**0.901**
+ SMOTE	75%	64%	0.69	0.897

Row highlighted in green is the sweet spot — threshold tuning on top of class weights.

468

of 600 defaults now caught — up from 0

Recall went from 0% to 78% with three configuration changes. No new model. No new data.

THE TRADE-OFF

Accuracy Drops. Everything That Matters Goes Up.

94% → 87%

Accuracy
(down 7 points)

0% → 78%

Recall
(up 78 points)

0.891 → 0.901

AUC
(barely moved)

THE TRADE-OFF IS WORTH IT

AUC barely moves — the model's underlying ability to rank borrowers was always there. What changed is how we translated that ranking into a decision. That's the lesson: the fix wasn't a smarter model, it was using the model you already had more intelligently.

SECTION 03

Naming the Blind Spot

THE UNCOMFORTABLE QUESTION

What Does Your Model Actually Know?

A supervised model learns from labels. It gets very good at recognizing patterns it has already seen labeled as fraud.

Its ceiling is the label.

It cannot catch a pattern that has never been labeled as fraud — no matter how well-tuned the threshold, no matter how balanced the classes.

What happens the day a genuinely new fraud pattern shows up — one that looks nothing like any of your 600 labeled defaults?

SECTION 04

A Different Question

THREE UNSUPERVISED METHODS

No Labels. No Target Variable. Just: What's Different?

METHOD 1

Isolation Forest

Finds borrowers that are easy to isolate globally — unusual across the entire portfolio.

METHOD 2

Local Outlier Factor

Finds borrowers unusual relative to their local neighbors — catches anomalies hidden within subgroups.

METHOD 3

One-Class SVM

Learns a boundary around normal borrowers and flags anything that falls outside it.

LIVE DEMO

Running Isolation Forest, LOF, and One-Class SVM on all 24,977 LendingClub borrowers — with bad_loan set completely aside.

[switch to screen share]

SECTION 05

The Payoff

AGREEMENT ANALYSIS

When Two Independent Methods Agree

14.3%

Flagged by BOTH
453 borrowers

10.8%

Flagged by ONE only
1,592 borrowers

13.2%

Flagged by NEITHER
22,932 borrowers

13.0%

Portfolio average
all 24,977 borrowers

THE INSIGHT

Borrowers flagged by only one method sit below the portfolio average — often benign outliers. Borrowers flagged by both methods exceed it. When two methods built on completely different logic agree, that convergence is a signal worth investigating.

SECTION 06

Unusual ≠ Risky

4

of 20

Of the 20 most anomalous borrowers in the entire portfolio — the ones the models were most confident about — only 4 actually defaulted.

Unusual is not the same statement as risky.

THREE CATEGORIES, SAME FLAG

What the Analyst Has to Determine

RISKY	**Genuinely risky**	Income $14,208 · DTI 65.12 · Rate 30.89% · FICO 664 · Bad loan: YES. Every feature signals distress — the model was right to flag this one.
BENIGN	**Benign outlier**	Income $3,000,000 · DTI 2.39 · Rate 7.21% · FICO 749 · Bad loan: no. Ultra-wealthy borrower, modest loan, prime rate. Unusual, not risky.
DATA	**Data quality issue**	Income $1,800 · DTI 999 · Rate 30.84% · FICO 729 · Bad loan: no. A DTI of 999 isn't real — this record needs investigation, not a credit decision.

SEVERITY TIERS

Not All Anomalies Get the Same Response

Extreme (top 1%)	~250	Immediate manual review
High (top 2%)	~250	Priority review queue
Moderate (top 5%)	~750	Enhanced monitoring
Borderline (top 10%)	~1,250	Watch list
Normal	~22,500	Standard processing

SECTION 07

The Framework

BRINGING IT TOGETHER

A Two-Layer Architecture

LAYER 1 — SUPERVISED

Catches known patterns

Class weights, tuned thresholds, labeled fraud history. Fast, precise, blind to anything it hasn't seen before.

LAYER 2 — UNSUPERVISED

Catches the unknown unknowns

Isolation Forest, LOF, One-Class SVM. No labels needed — flags what doesn't fit, tiered by severity, reviewed by a human.

The takeaway

Fixing accuracy fixes what you can see. A second layer catches what you can't.

Supervised models are only as good as their labels. Unsupervised anomaly detection is how you find out what you don't know you're missing — before it becomes a pattern with a name.

Questions?

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Full code for both tutorials this talk is built on is available at james-mott.jimmott52.workers.dev/#tutorials